SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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Technical & Tools Declaration

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1. Team & Project Overview

Team Name	418 I'm a Teapot
Project / Solution Name	AI Career Navigator
Name (Leader)	CHEN XIANGYU
Email & Phone (Leader)	CufoFlux@outlook.com +60173568532
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://aicareer.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/Sleepy-Tianhao/SDC_Hackathon_418-I-m-a-Teapot_Submission

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Our project is built on a simple but efficient client-server architecture. The frontend collects user inputs (like target roles and existing skills) and sends a JSON request to the FastAPI backend. To prevent AI hallucination and save token costs, the backend doesn't blindly ask the AI. Instead, it uses Pandas and Openpyxl to dynamically search through 8 real-world Malaysian labor market datasets (CSV/XLSX files) stored on the server. By extracting only the most relevant data rows, the backend creates a highly accurate context. This data is then combined with a custom prompt and sent to the DeepSeek API. The LLM processes this factual grounding and returns a structured JSON containing personalized reskilling pathways and SDG mappings, which the frontend renders. The entire app is deployed on Replit.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
JavaScript	Enhance the interactivity and dynamism of web pages	ECMAScript Specification License
Python 3	Backend logic and API development.	Python Software Foundation License Version 2
CSS	Add styles to HTML or XML documents, including controlling colors, fonts, layouts, spacing, animations, and other visual presentations, in order to separate structure from style in web pages, enhancing maintainability and reusability.	MIT License
HTML	Create a Web Page	Creative Commons Attribution 4.0 International License

Name & Version	Purpose in Project	License
FastAPI	Modern Frameworks for Web APIs / Backend Services	MIT License
Pandas	Data processing and RAG retrieval from CSV files.	BSD 3-Clause License
Openpyxl	Parsing Excel (XLSX) datasets.	MIT License

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
DeepSeek: https://api.deepseek.com	Analyze user data and generate feedback.	paid

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit: https://your-project.replit.app/	Deploy and host the project	sponsor-provided

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
DeepSeek-V4.1-Flash-1017	Hangzhou Deep Search Artificial Intelligence Basic Technology Research Co., Ltd	pre-trained	MIT License
Deepseek-chat	Hangzhou Deep Search Artificial Intelligence Basic Technology Research Co., Ltd	pre-trained	MIT License

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Malaysia Graduate Employment Rate (MoHE 2024)	https://www.mohe.gov.my	public	Open Data	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
National & State Labour Force Statistics (DOSM)	https://data.gov.my	public	Open Data	N
Salary by Qualification & Industry Distribution	https://data.gov.my	public	Open Data	N
Global Labour Market Trends (2024)	https://www.ilo.org	public	Open Data	N
AI & Future Skills Signals	Synthesized from public reports	synthetic	N/A	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit	Deploy and host the project	minor
DeepSeek Harness DeepSeek-V4.1-Flash-1017	Develop the frontend and backend of the project & Generative AI briefing	major
Deepseek-chat	Structured JSON Call	major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

While we used AI assistants to speed up coding, we maintained strict human oversight. Every FastAPI endpoint was manually tested locally before deployment. To prevent AI hallucinations in the final product, we designed a RAG (Retrieval-Augmented Generation) pipeline. The AI is strictly constrained to analyze only the real data we parsed from official Malaysian datasets, rather than relying on its pre-trained memory. All output is validated against a strict JSON schema.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	$25	Deploy and host the project

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is essential for providing a live, public URL for the judges to interact with our prototype. It serves our FastAPI backend and frontend seamlessly. However, our code is built with standard Python and is entirely platform-agnostic. If Replit were to become unavailable, we could migrate the entire project to another cloud provider like Render or AWS within a few hours just by updating the environment variables and deployment scripts.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

We take user privacy seriously. Our application does not require account registration, and we do not store any personal data (like names, emails, or IPs) in a persistent database. The user's skills and target roles are processed in memory during the session and discarded immediately after. To generate the AI pathway, this non-personal data and the relevant labor statistics are sent to the DeepSeek API via a secure HTTPS connection. Furthermore, we have implemented CORS (Cross-Origin Resource Sharing) middleware to ensure secure communication between our frontend and backend.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Limitations: The main limitation is our dependency on the third-party DeepSeek API. Network latency can cause generation to take 3-5 seconds, and we are subject to their rate limits. Additionally, our RAG search relies on keyword matching; if a user inputs highly niche skills not present in our CSV files, the AI has limited local data to work with.
Mitigations: To handle API failures, we implemented robust try...except blocks in our FastAPI backend so the server won't crash. If the API times out, it returns a clean error message. If no CSV data matches the user's query, our prompt design falls back to general AI reasoning, ensuring the user always receives a helpful response without breaking the app.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

[Date]